EE426 / 626 Microprocessor Applications

Brief History of the Intel 8051 (MCS-51) Chipset



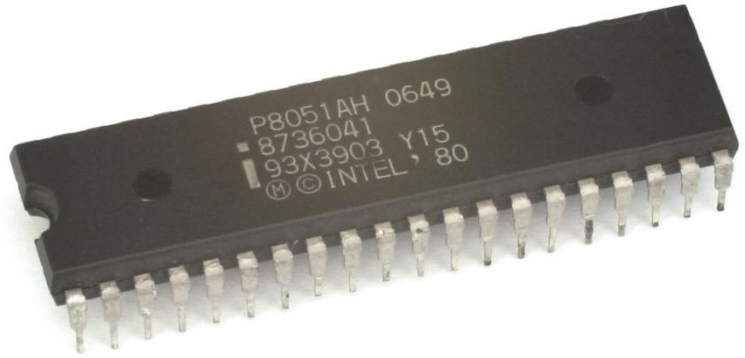
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Intel 8051/52

MCS-51 Family

1980 Intel releases 8051 (successor to 8048)

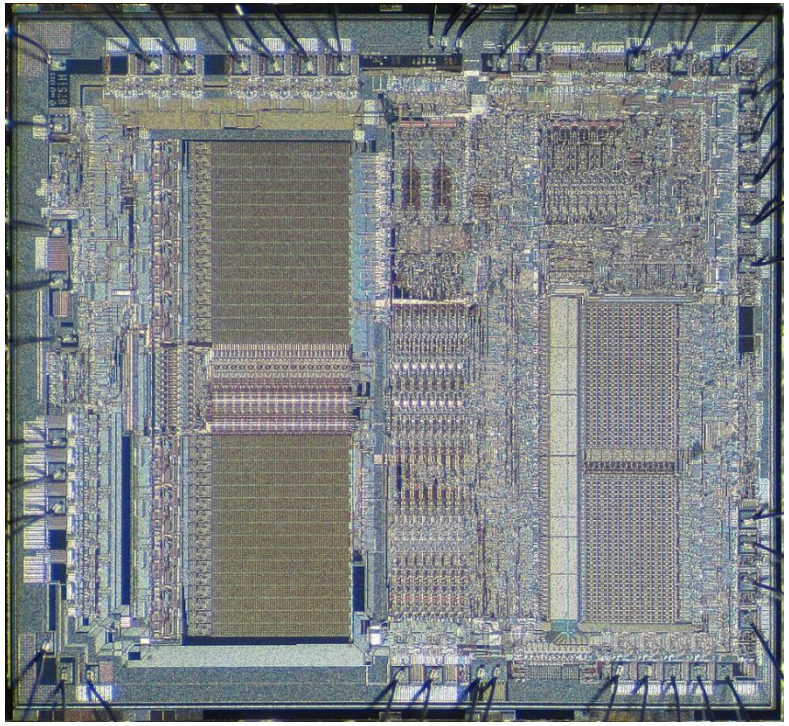
- Might Be Most Famous Microcontroller (MCU)
- 3 μm HMOS (depletion-mode NMOS) → 2 μm
- 40-pin PDIP/CDIP
- Harvard Architecture
  - 16-bit address (64 kB PMEM)
  - 16-bit address (64 kB XRAM)
  - 128 or 256 bytes on-chip data RAM
- 4 x 8-bit I/O Ports
- 2 or 3 x 16-bit Timer / Counter and 1 or 2 UARTs
- Single +5 V_{CC} Supply with 12 MHz clock
- Interrupt support with banked context registers
- 90+ Instructions with Multiply and Divide (!)



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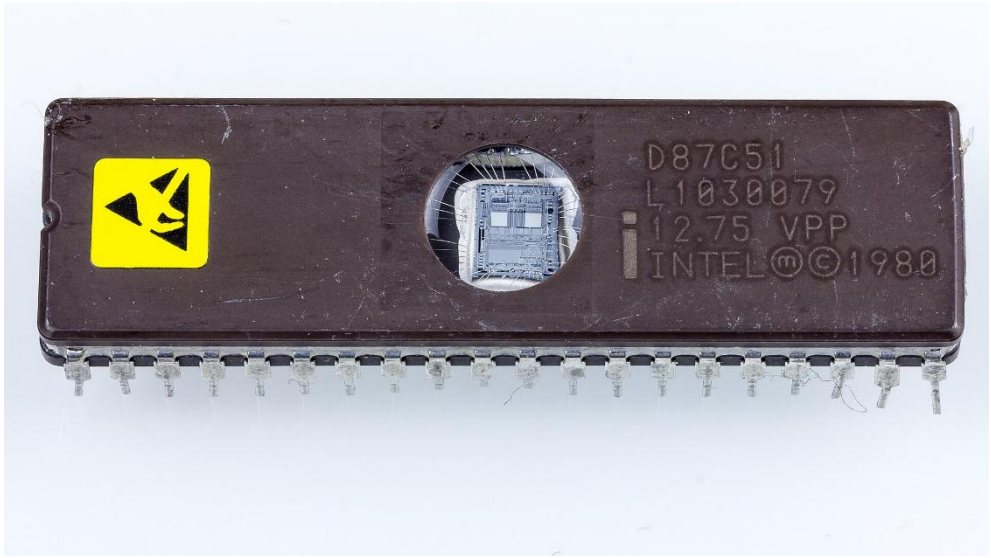
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Intel D87C51 with 4 kB x 8 EPROM



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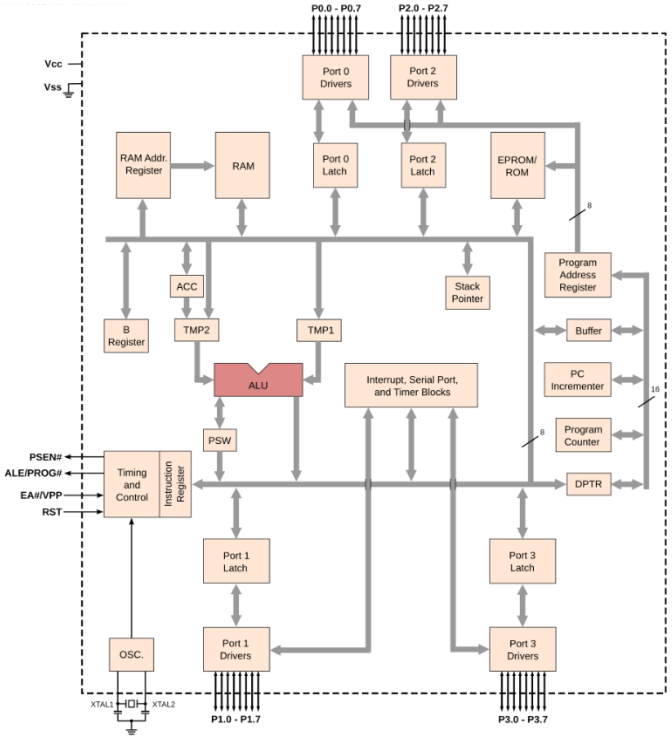
Intel D87C51 with 4 kB x 8 EPROM



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Intel 8051 Microarchitecture



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Cypress CY7C68xxx

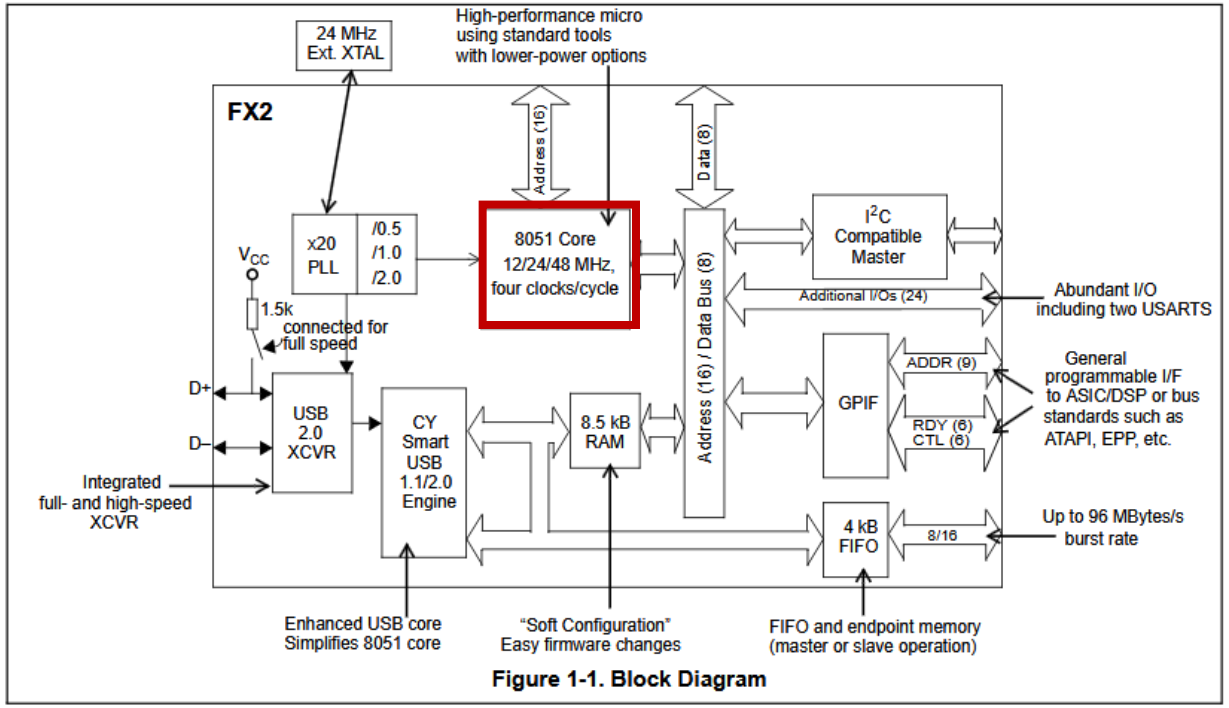
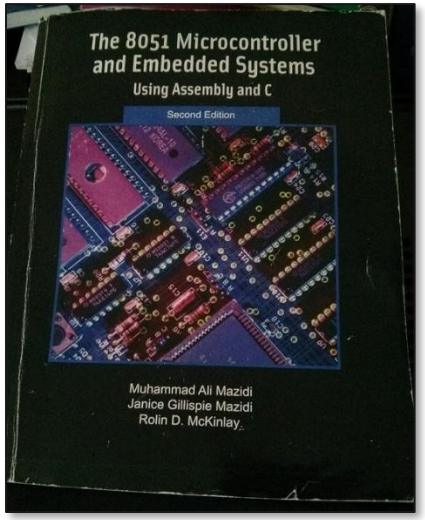
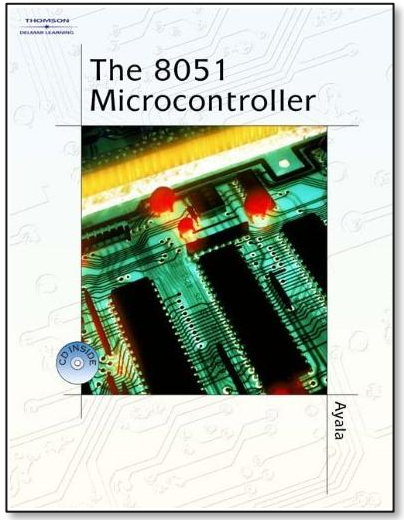


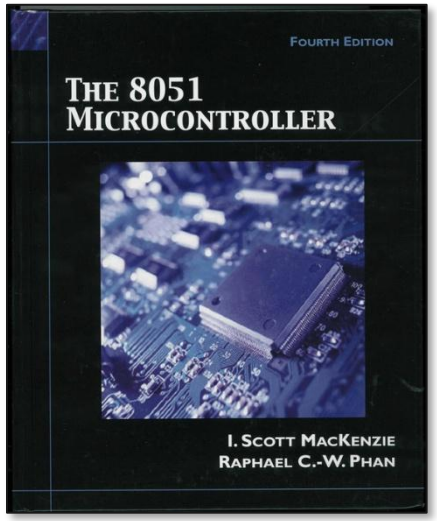
Figure 1-1. Block Diagram

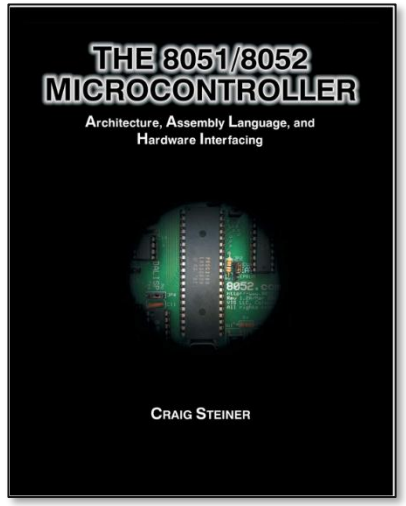
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https://www.keil.com/dd/docs/datashts/cypress/cy7c68xxx_ds.pdf

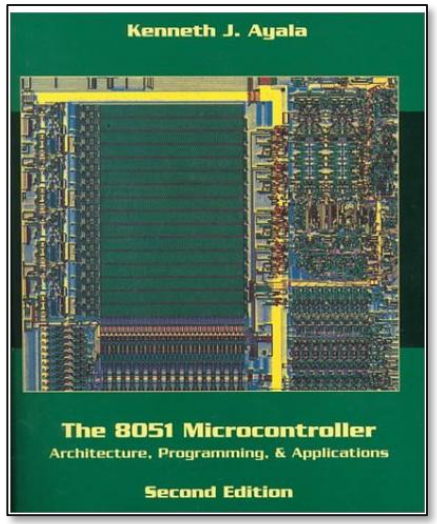
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MCS-51 family:

- 8031 – 8-bit Control-Oriented Microcontroller
- 8032 – 8-bit Control-Oriented Microcontroller
- 8044 – High Performance 8-bit Microcontroller
- 8344 – High Performance 8-bit Microcontroller
- 8744 – High Performance 8-bit Microcontroller
- 8051 – 8-bit Control-Oriented Microcontroller
- 8052 – 8-bit Control-Oriented Microcontroller
- 8054 – 8-bit Control-Oriented Microcontroller
- 8058 – 8-bit Control-Oriented Microcontroller
- 8351 – 8-bit Control-Oriented Microcontroller
- 8352 – 8-bit Control-Oriented Microcontroller
- 8354 – 8-bit Control-Oriented Microcontroller
- 8358 – 8-bit Control-Oriented Microcontroller
- 8751 – 8-bit Control-Oriented Microcontroller
- 8752 – 8-bit Control-Oriented Microcontroller
- 8754 – 8-bit Control-Oriented Microcontroller
- 8758 – 8-bit Control-Oriented Microcontroller

MCS-151 family:

- 80151 – High Performance 8-bit Control-Oriented Microcontroller
- 83151 – High Performance 8-bit Control-Oriented Microcontroller
- 87151 – High Performance 8-bit Control-Oriented Microcontroller
- 80152 – High Performance 8-bit Control-Oriented Microcontroller
- 83152 – High Performance 8-bit Control-Oriented Microcontroller

MCS-251 family:

- 80251 – 8/16/32-bit Microcontroller
- 80252 – 8/16/32-bit Microcontroller
- 80452 – 8/16/32-bit Microcontroller
- 83251 – 8/16/32-bit Microcontroller
- 87251 – 8/16/32-bit Microcontroller
- 87253 – 8/16/32-bit Microcontroller



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